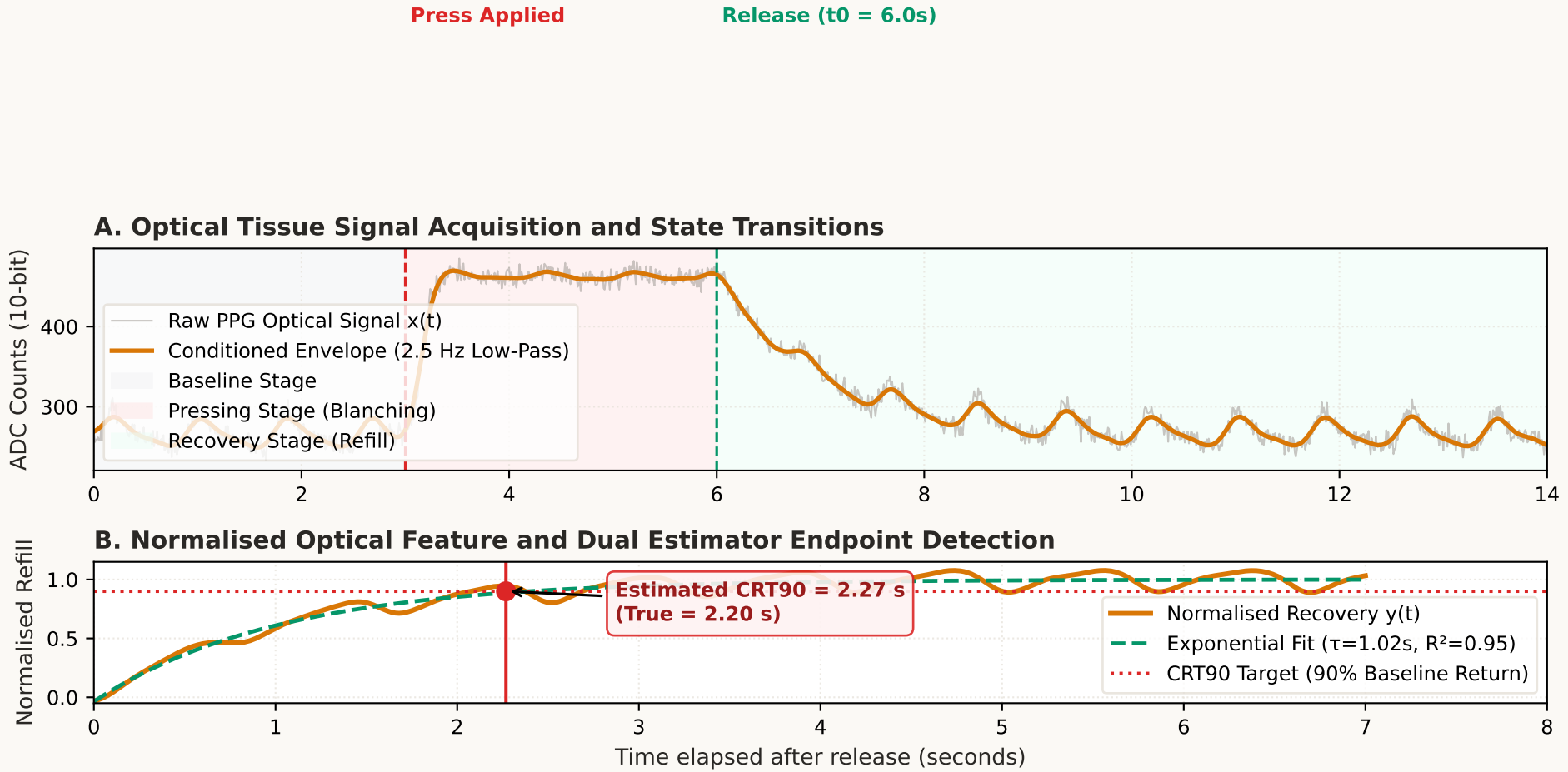

Press Applied
Release (t0 = 6.0s)
A. Optical Tissue Signal Acquisition and State Transitions
ADC Counts (10-bit)
Raw PPG Optical Signal x(t)
Conditioned Envelope (2.5 Hz Low-Pass)
Baseline Stage
Pressing Stage (Blanching)
Recovery Stage (Refill)
B. Normalised Optical Feature and Dual Estimator Endpoint Detection
Normalised Refill
Estimated CRT90 = 2.27 s
(True = 2.20 s)
Normalised Recovery y(t)
Exponential Fit (τ=1.02s, R²=0.95)
CRT90 Target (90% Baseline Return)
Time elapsed after release (seconds)